

Final Exam – Project

Equilibrium Prices and Quantity at Current State Tax Rate:

State Tax rate: 0.40

Demand: $D(p_B) = 6 - .36p_B$

Supply: $S(p_S) = 4.5 + .23p_S$

$p_B = p_S + .40$

$$6 - .36(p_S + .40) = 4.5 + .23p_S \rightarrow 5.856 - .36p_S = 4.5 + .23p_S$$

$$1.356 = .59p_S \rightarrow p_S = 2.298 \rightarrow p_B = 2.298 + .40 = 2.698$$

$$D(2.698) = 6 - .36(2.698) = 5.029$$

$$S(2.298) = 4.5 + .23(2.298) = 5.029$$

Price Buyer's Pay: 2.698/gal

Price Seller's Receive: 2.298/gal

Equilibrium Quantity: 5.029b gallons

1.

Equilibrium Prices and Quantity at Proposed Zero State Tax Rate:

State Tax rate: 0

Demand: $D(p_B) = 6 - .36p_B$

Supply: $S(p_S) = 4.5 + .23p_S$

$p_S = p_B$

$$6 - .36p_B = 4.5 + .23p_S \rightarrow 6 - .36p_B = 4.5 + .23p_B \rightarrow 1.5 = .59p_B$$

$$p_B \text{ and } p_S = 2.542$$

$$D(2.542) = 6 - .36(2.542) = 5.085$$

$$S(2.542) = 4.5 + .23(2.542) = 5.085$$

Price Buyer's Pay: 2.542/gal

Price Seller's Receive: 2.542/gal

Equilibrium Quantity: 5.085b gallons

An elimination of the state tax rate results in a .156 cent reduction (2.698 – 2.542) in the price buyer's pay for gasoline.

2.

Consumer Expenditure with State Tax:

Quantity of gasoline purchased in gallons: 5,029,000,000

State tax included price per gallon: \$2.698

Total consumer expenditure: \$13,568,242,000

Consumer Expenditure without State Tax:

Quantity of gasoline purchased in gallons: 5,085,000,000

State tax included price per gallon: \$2.542

Total consumer expenditure: \$12,926,070,000

Difference in Consumer Expenditure:

$$\$13,568,242,000 - \$12,926,070,000 = \$642,172,000$$

The proposed removal of state tax on gasoline sales would save consumers roughly 642 million dollars annually, 1.357 billion dollars less than, or only 32.1% of, the savings argued by the politician.

3.

Producer Revenue with State Tax:

Quantity of gasoline sold in gallons: 5,029,000,000

State tax included income per gallon: \$2.298

Total producer revenue: \$11,556,642,000

Producer Revenue without State Tax:

Quantity of gasoline sold in gallons: 5,085,000,000

State tax included income per gallon: \$2.542

Total producer revenue: \$12,926,070,000

Difference in Producer Revenue:

$$\$12,926,070,000 - \$11,556,642,000 = \$1,369,428,000$$

The proposed removal of state tax on gasoline sales would increase producer revenue by roughly 1.37 billion dollars annually.

4.

Consumers: Eliminating state taxes on gasoline sales provides gains for consumers totaling over 642 million dollars. Moreover, though consumers are spending less overall, 56 million additional gallons of gasoline are consumed indicating that, without the state tax, consumers will spend less on gas while receiving more. Within the specific context of the market for gasoline, consumers are clear winners.

Producers: Eliminating state taxes on gasoline sales provides gains to producers totaling over 1.36 billion dollars. While increased demand will require increasing output in the production process, producers receive over twice the financial benefit than consumers. Within the specific context of the market for gasoline, producers are also clear winners.

State Government Revenues: Somewhat intuitively, the negative effects on state tax revenues lead to the largest losses being incurred by state government revenues. The quantity of gasoline consumed under the current taxation structure is 5.029 billion gallons. With a .40 cent per gallon tax, the state earns roughly 2.012 billion dollars annually from the sale of gasoline. Eliminating the state tax on gasoline sales would mean that programs currently funded by these revenues would either need alternative funding or to be eliminated alongside the tax that pays for their continued operation.

5.

Looking purely at efficiency in the gasoline market, eliminating the gasoline tax is a good idea. The tax introduces deadweight loss that results in a loss of welfare for society. The total value of that loss can be calculated as $\frac{1}{2}$ the difference between the per gallon consumer price (\$2.698) and producer income (\$2.298), multiplied by the difference between gallons consumed at equilibrium with (5.029b) and without the tax (5.085b). The lost welfare to society totals 11.2 million dollars ($.5(.40) * (5.085 - 5.029) = .0112 \text{ billion}$). Eliminating the tax also eliminates the losses arising from the inefficiency in the market that is created by their existence. Therefore, when exclusively examining economic efficiency in the gasoline market, eliminating the tax restores that lost welfare back to society.

6.

Taking a broader view, I do not believe that eliminating the gasoline tax is a good idea. North Carolina's gasoline tax revenue is used to support a wide variety of transportation and environmental programs. Specifically, the funds are largely allocated to road construction and maintenance with smaller percentages going toward leaking petroleum and water-quality cleanup efforts, and small dedicated programs for boating safety and dredging of navigation channels, as outlined in [Part 7 of Article 36C](#).

While consumers and producers benefit from reduction in the gas tax, those benefits may be far outweighed by the negative externalities created by a lack of funding for critical infrastructure and environmental programs. The gain in efficiency (difference in deadweight loss) is only 11.2 million dollars, whereas the lost government revenue exceeds 2 billion. Since these revenues are allocated not only to maintenance of the infrastructure which benefits gasoline producers and consumers, but also to programs which benefit everyone, society at large is harmed by this reduction. Furthermore, environmental programs have the potential to create long-term health benefits which, while impactful, may be difficult to accurately measure.

For these reasons, NC should not eliminate the state gasoline tax. In fact, I believe that raising the tax could help to fund additional programs such as rural EV charging infrastructure, investment in biofuels, and additional repairs and maintenance for roadways in addition to furthering the missions of existing environmental and safety programs. Valuation of the benefits of these programs would be essential to weighing any additional deadweight loss created by a higher tax rate.